obtain an estimate of how small b might be, and ultimately obtain the size of the nucleus—and the size of the nucleus turned out to be 10^{-5} times smaller than the atom! This was the way that it was discovered that nuclei exist.

3-3 The fundamental rocket equation

Now, the next problem I want to talk about is completely different: it has to do with rocket propulsion, and I'm going to take a rocket floating around in empty space first—forgetting all about gravity, and so on. The rocket's built to hold a lot of fuel; it's got some kind of engine by which it squirts fuel out the back—and from the point of view of the rocket, it's always squirting it out at the same speed. It doesn't turn on and off; we start it, and it just keeps squirting stuff out the rear end until it runs out. We'll suppose that the stuff is squirted out at a rate of μ (that's mass per second), and that it goes out at velocity u. (See Fig. 3-9.)

You might say, "Aren't those the same thing? You know the mass per second; isn't that the velocity?"

No. I can dump a certain amount of mass per second by taking a great big lump of stuff and putting it quietly out each time, or I can take the same mass and *throw* it out each time. So, you see, they're two independent ideas.

Now, the question is, how much velocity will the rocket accumulate after a time? Suppose, for instance, that it uses up 90 percent of its weight: that is, when it's finished using all its fuel the mass of the shell that's left is one-tenth as great as the mass of the whole thing loaded before it started. What speed will the rocket acquire?

Anybody in his right mind would say that it is impossible to get any faster than the speed u, but that's not true, as you'll see in a moment. Maybe you'll say that's perfectly obvious; well, all right. But it is, in fact, true for the following reason.

Let's look at the rocket at any moment, moving at any speed at all. If we move along with the rocket and watch for a time Δt, what do we see? Well, there's a certain mass Δm that goes out—which is, of course, the rocket's

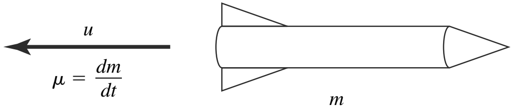

FIGURE 3-9 Rocket with mass m, ejecting fuel at rate $\mu = dm/dt$ with velocity u.